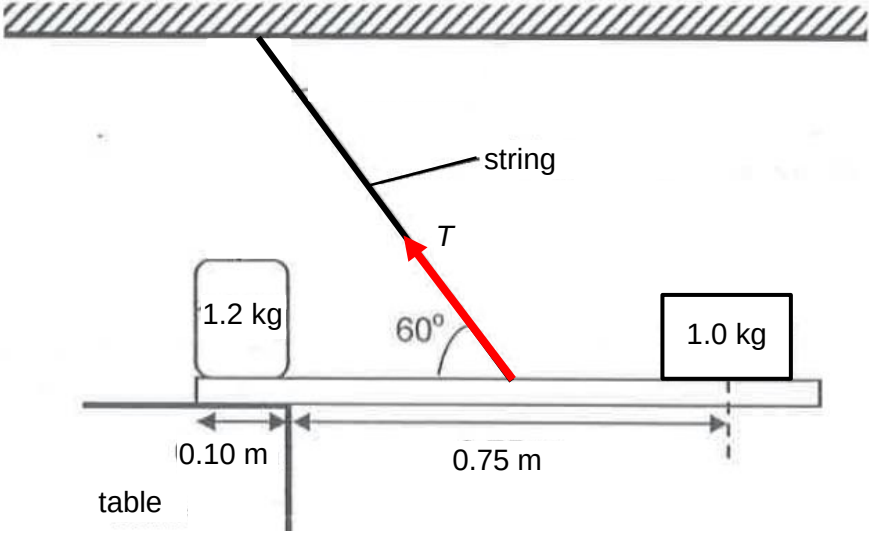


	$a = 15.5 \text{ m/s}^2 = 20 \text{ m/s}^2$	answer
3(a)(i)	The moment of a force about a pivot is the <u>product</u> of the <u>force</u> and the <u>perpendicular distance from the pivot to the line of action of the force</u> .	[1]
3(a)(ii)	Let N be the number of 5.0 g mass needed for equilibrium. Taking moments about the edge of the table, anti-clockwise moments = clockwise moments $(1.2)g(0.05) = (0.11)g(0.40) + N(0.005)g(0.75)$ $N = 4.3$ Since N is a whole number, the maximum number is 4.	[1] correct substitution [1] correct answer
3(b)(i)	 The diagram shows a horizontal beam resting on a table. A 1.2 kg mass is placed on the beam, 0.10 m from the left edge. A 1.0 kg mass is placed on the beam, 0.75 m from the right edge. A string is attached to the ceiling and the beam, making a 60-degree angle with the beam. The tension in the string is T.	[1] for T

3(b)(ii)	Taking moments about the edge of the table, anti-clockwise moments = clockwise moments $(1.2)g(0.05) + T \sin 60^\circ(0.40) = (0.11)g(0.40) + (1.0)g(0.75)$ $0.346T = 7.2$ $T = 20.8 \text{ N} = 21 \text{ N}$	[1] correct substitution [1] correct answer
4(a)	Since the direction of motion of the body is always changing its	[1]